

To:

The Editors of *PLOS Computational Biology*

Dear Editors of *PLOS Computational Biology*,

Please consider our manuscript, “Stable and recurrent deformation subspaces preserve pelvic mechanical function across anatomical variability,” for publication as a Research Article in *PLOS Computational Biology*.

A central problem in population biomechanics is how to compare mechanical deformation pathways across anatomically variable individuals. Standard modal analyses typically compare eigenmodes by rank, implicitly assuming that the same rank represents the same biological function in every subject. We show that this assumption can fail in crowded regions of the mechanical spectrum: small anatomical perturbations can exchange neighbouring mode labels even when the underlying functional deformation capacity remains stable.

To address this problem, we developed a basis-invariant subspace framework for population-scale biomechanical comparison and applied it to 278 CT-derived subject-specific pelvic finite-element models. We identify two distinct organisational regimes. Modes 1–3 form a gap-protected, order-stable mechanical backbone that dominates habitual standing. At higher ranks, mechanical function is instead organised within compact recurrent subspaces whose individual mode labels can exchange between subjects.

The clearest example is the recurrent rank 9–10 subspace, present in 54.7% of the cohort. During label exchange, AP-inlet coupling assigned to an individual rank changes dramatically, whereas the corresponding basis-invariant capacity of the shared two-dimensional subspace remains remarkably stable. The same subspace is selectively recruited by standardized internal-ring expansion loads. Thus, a mode-by-mode comparison can suggest functional disruption where the mechanically relevant subspace has in fact changed very little.

We further connect this population-level behaviour to spectral mechanics using a minimal analytical model of near-degenerate mode veering and quantify whole-pelvis deformation using a basis-invariant kinematic decomposition. Together, these analyses show that spectral rank, deformation pathway, and biological function are distinct objects, and provide a general computational strategy for comparing functional mechanics across variable anatomy.

We believe the manuscript is particularly suited to *PLOS Computational Biology* because the methodological problem extends beyond pelvic biomechanics. Near-degenerate spectra and unstable eigenvector labels occur broadly in computational models of biological structures. The proposed framework provides a reproducible way to identify population-stable mechanical programs without assuming one-to-one conservation of individual modes.

The manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have reviewed and approved the manuscript and declare no competing interests. Computational analysis code and derived data required to reproduce the reported analyses will be made available through the study repository in accordance with journal policy.

Thank you for your consideration.

Sincerely,

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